Issue	**01**
Factory ID	**25796**
Date of Issue	**29-Jul-2025**

Ebrahim Textile Mills

PLOT HX2,ZONE 2LANDHI INDUSTRIAL AREA KARACHI
Karachi
Sindh Pakistan

Latitude: [24.840655], longitude: [67.239885]
Date of inspection: 13-May-2025

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 13-05-2025 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 13-May-2025, Shahzeb Anis and Zeeshan Haider on behalf of the Pakistan Accord carried out an initial electrical inspection for Ebrahim Textile Mills, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Syed Hurr (Sr Manager Compliance), Mr. Aqeel Mohsin (G.M Engineering), Mr. Jamil Hussain (Electrical Manager), Mr Khobaib Javaid (Manager Civil) and other staff members of Ebrahim Textile Mills (Pvt) Ltd. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection.

The factory operates with one general shift and three shifts for processing only. Approximately 1380 workers (peak occupants) operate in the single shift of the factory. Factory employs 1380 workers out of which approximately 2% employees are female.
The factory comprises 6 buildings owned by Ebrahim Textile Mills (Pvt) Ltd. During the inspection, the buildings highlighted in Figure 1.1 were surveyed, with the site boundary indicated in red. The building highlighted with green in figure 1.1 was not inspected during the initial inspection. This building shall be included in future follow-up inspections.

- **Building 1**- a 1-storey and 2-storeys (Ground + 1). The building is used for, empty (1st floor) and admin HR, Godowns, chemical Godown, warehouse, workshops, carton Godowns (ground floor). The building was constructed in 1991.
- **Building 2**- a 1-stoery and 2-storeys (Ground + 1). The building is used for, marketing office (1st floor) and digital printing, embroidery, main store, local folding, masjid, workshops (ground floor). The building was constructed in 1991.
- **Building 3**- a 1-storey (Ground only). The building is used for greige department and engineering offices (ground floor). The building was constructed in 1991.
- **Building 4**- is a 1-storey (Ground only) . The building is used for stitching and warehouse (ground floor). The year of construction of the building is 1991.
- **Building 5**- is consists of 5 different structures which are interconnected at ground level such that you can move freely between them. The building was constructed in 1991 and a part of the building extended horizontally in 2013 and vertically in 2022.
  - Structure 1 – 1-storey (Ground only) used for processing, folding, storage (ground floor).
  - Structure 2 – 1-storey (Ground only) used for washing and storage (ground floor).
  - Structure 3 – 1-storey (Ground only) used for bleaching and labs (ground floor).
  - Structure 4 - 5-storey (G+M+3) is used for cutting (roof), stitching (3rd and 2nd floor), engraving (1st floor), FG storage (mezzanine) and processing and folding (ground floor).
  - Structure 5 - 2-storeys (Ground +1) used for quilting (1st floor) and cutting (ground floor).
- **Building 6**- is consist of 2 different structures. The year of construction of the building is 1991 (Steel Portal Frame) in 2018 and 2021.
- **Structure 1**- a 1-storey (Ground only). The building is used for local processing (ground floor).

- **Structure 2**- a 1-storey (Ground only). The building is used for local processing (ground floor).

A summary of main concerns identified on-site during the assessment (non-exhaustive) that should be actioned are as follows:

• **Hot Spots:** The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires.
• **Fire Alarm & Detection System (FADS):** The fire alarm and detection system was found to be non-listed and non-addressable, with major deficiencies including inadequate detector coverage, use of non-fire-rated cables, and faulty sounders and visual indicators. Critical areas lacked detectors entirely, and conventional panels without proper system integration were in use, posing serious safety risks and non-compliance with fire safety standards.
• **Emergency lighting:** It was noted that several emergency lights did not comply with building standards. Additionally, there was a lack of emergency lighting coverage as per standard evacuation requirements

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.2. Site plan (Satellite view) Red dashed line indicates site boundary, and Yellow dashed line indicates building boundaries, highlighted (blue) area indicates buildings inspected and highlighted (green) will inspect in future follow-up inspection.

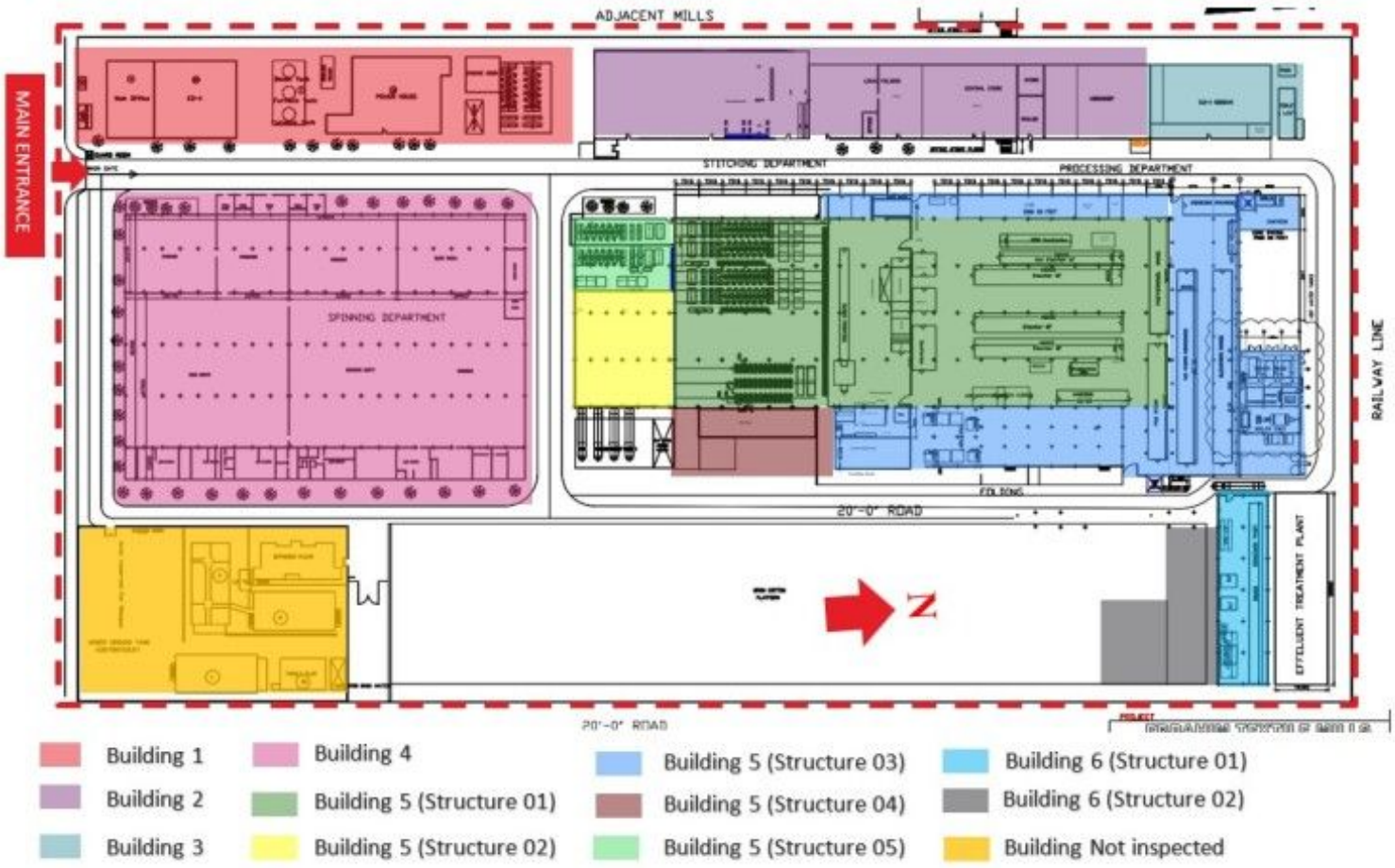


Figure 1.3. Site layout. Red dashed line indicates site boundary.

2.1 Building Use

The factory comprises of six main buildings with details as below:

Building 1- (Ground + 1)
L00 - Ground floor is used for admin HR, Godowns, chemical Godown, warehouse, workshops, carton Godowns.
L01 - First floor is empty

Building 2- (Ground + 1)
L00 - Ground floor is used for digital printing, embroidery, main store, local folding, masjid, workshops
L01 - First floor is used for marketing office

Building 3- (Ground only)
L00 - Ground floor is used for greige department and engineering offices

Building 4- (Ground only)
L00 - Ground floor is used for stitching and warehouse

Building 5- is consists of 5 different structures
Structure 1 – (Ground only)
L00 - Ground floor is used for processing, folding, storage
Structure 2 – (Ground only)
L00 - Ground floor is used for washing and storage
Structure 3 – (Ground only)
L00 - Ground floor is used for HT, LT generator room, bleaching and labs
Structure 4 - (G+M+3)
L00 - Ground floor is used for processing and folding
M01 - Mezzanine is used for FG storage
L01 - First floor is used for engraving
L02 - Second floor is used for stitching
L03 - Third floor is used for stitching
L99 - Roof is used for cutting
Structure 5 - (Ground +1)
L00 - Ground floor is used for used for cutting (ground floor).

L01 - First floor is used for quilting

Building 6- is consist of 2 different structures.
Structure 1- (Ground only).
L00 - Ground floor is used for local processing
Structure 2- (Ground only).
L00 - Ground floor is used for local processing

2.2 Utility Source

The facility operates a self-generation power system via a Powerhouse equipped with three Gensets of sizes 1064 kW and 1169 kW (qty = 2) that operates on natural gas. The generated power than enters the adjacent LT Room at 400V. Further, the supply is distributed throughout the facility via cables and busways.

2.3 Backup source

This facility has 11kV supply from k Electric as backup source connected by a 1000kV transformer. This operates upon the loss of the mains power. Fire alarm panel and emergency lights were at built-in battery power.

3. Documentation

This table shows the details of the documentation provided by the factory.

S.NO	Description	Availabe (Y/N)	Created Date	Status / Remarks
1	As-Built Layout Drawing	No		
2	Single Line Diagrams	No		
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	No		

Table 1: Documentations

4. Electrical Safety Measures

During our electrical site inspection of Ebrahim Textile Mills, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Ebrahim Textile Mills. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a trained professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Diagrams & Drawings

5.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202505-017559

Timeline	Remediation Plan
Medium Term FIN-0022752	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term FIN-0022753	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

5.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202505-017555

Timeline	Remediation Plan
Medium Term FIN-0022746	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term FIN-0022747	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

5.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202505-017557

Timeline	Remediation Plan
Medium Term FIN-0022748	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term FIN-0022749	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

6. DBs & Elec Protection Systems

6.1 Circuit breakers

This section refers to the entire facility.

During the survey the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

i. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes. This issue was specifically observed at DBP 01 in Building 05, where a 160A circuit breaker was connected using a 2.5 sqmm cable, and at DBP 05 in Building 05, where a 250A circuit breaker was connected using a 150 sqmm cable, both posing significant safety and compliance risks due to undersized conductors.
ii. Improperly mounted circuit breakers were observed in various areas of the factory. These breakers were installed without proper support or in non-standard positions, which can result in mechanical stress, loose connections, or unstable operation.
This issue was specifically observed at DBP 05 and DBP 04 in Building 05, the processing lab and process lab in Building 05, the outer area of the grey storage in Building 03, and the ground floor panel in the HR building, where circuit breakers were found mounted improperly.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202505-017547

Timeline	Remediation Plan
Medium Term FIN-0022608	i. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors. ii. All circuit breakers should be inspected and reinstalled properly using appropriate brackets and enclosures as per standard electrical installation practices to ensure safe and reliable operation.



Figure 2.1 - 160A circuit breaker connected with 2.5sqmm cable DBP 01 building 05



Figure 2.2 - 250A circuit breaker with 150sqmm cable at DBO 05 building 05



Figure 2.3 - Inappropriately mounted circuit breaker at DBP 05 in building 05

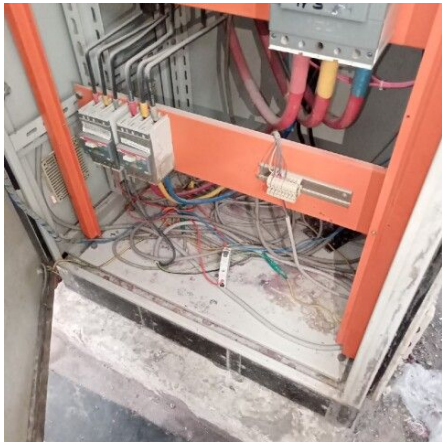


Figure 2.4 - Inappropriate mounted circuit breaker at DBP 04 building 05



Figure 2.5 - Inappropriate mounted circuit breaker at process lab building 05

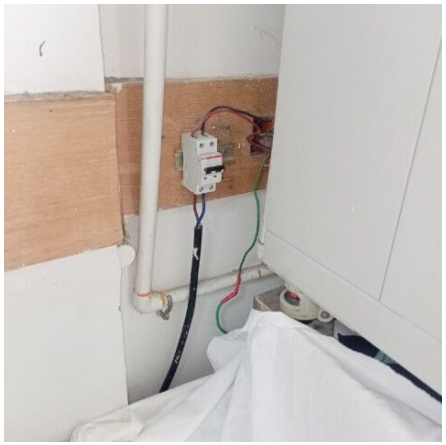


Figure 2.6 - Inappropriate mounted circuit breaker at processing lab building 05



Figure 2.7 - Inappropriate mounted circuit breaker at HR building ground floor panel



Figure 2.8 - Inappropriate mounted circuit breaker at outer area of grey storage building 03

6.2 Hot Spots

This relates to the entire facility.
During the inspection, Thermographic surveys were conducted on the electrical boards.
The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:
> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

All temperature-related assessments are based on Table 100.18 of ANSI Standard 2019.

This issue was specifically observed near the processing offices LDB, DB 07, and the AC breaker at Building 05.

It was also found that the Thermographic report was available at the site.

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment. F-202505-017549

Timeline	Remediation Plan

Immediate FIN-0022671	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term FIN-0022672	All circuits where the temperature was exceeding 20°C above ambient temperature, as illustrated in the example images below. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term FIN-0022673	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit thermography report to Pakistan Accord for information.



Figure 2.9 - Hot spot at AC breaker near processing

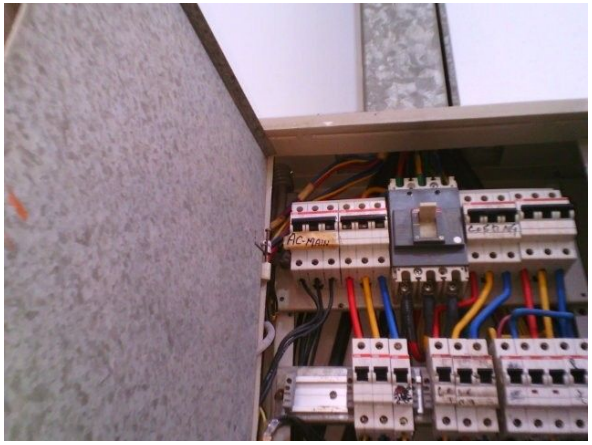


Figure 2.10 - Hot spot at AC breaker near processing offices at

offices at building 05

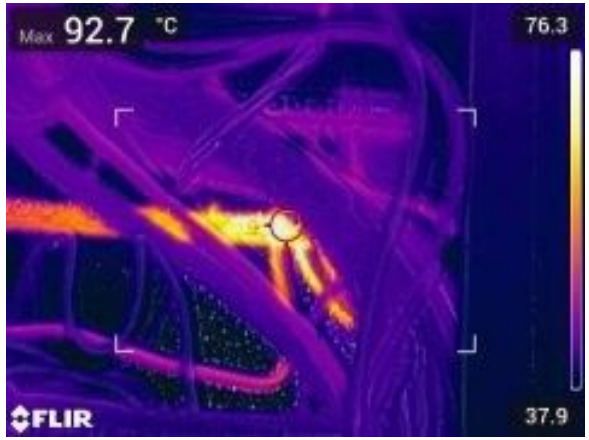


Figure 2.11 - Hot spot at cable termination near processing offices LDB building 05

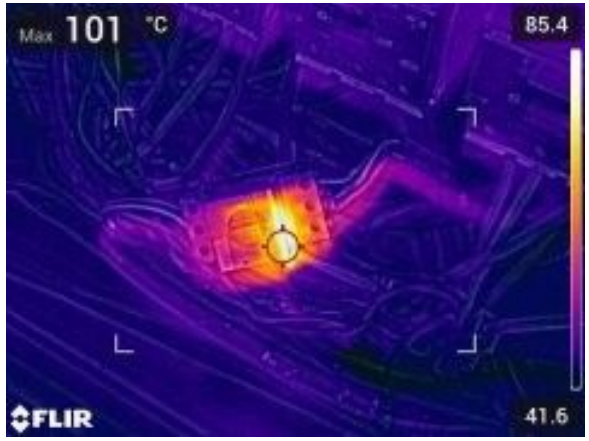


Figure 2.13 - Hot spot at MCCB in DB 07 building 05

building 05



Figure 2.12 - Hot spot at cable termination near processing offices LDB building 05

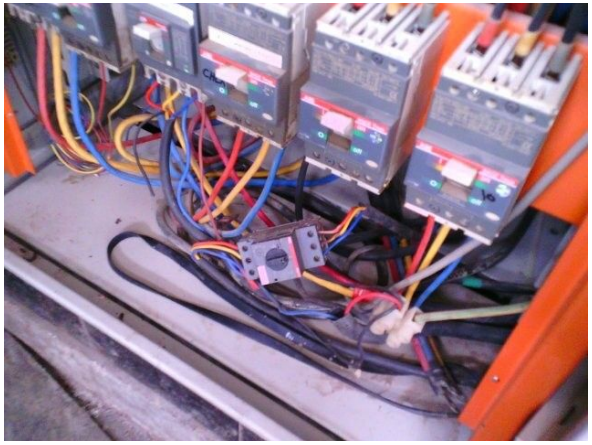


Figure 2.14 - Hot spot at MCCB in DB 07 building 05

6.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. There was no earthing system design and test report available in the facility.
ii. The earth pits were rusted and unsuitably located. This issue was specifically observed at the generator area, where earth pits were found without safety covers and the earth busbar was noted to be rusted and lacking proper safety enclosures.
iii. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables, and many electrical machines did not have proper earth connections. This issue was specifically observed at the grey storage panel in Building 03, where no earth connection was found.
iv. Equipotential bonding for the cable trays and metallic panel covers was missing. This issue was specifically observed at the distribution boards of Building 05 and at the lighting distribution board in the power house, where no equipotential bonding was found.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202505-017545

Timeline	Remediation Plan
Medium Term FIN-0022585	i. Design and evaluate the earthing system installation throughout the facility. Submit test reports and earth design layout to Pakistan Accord for information. ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54. iii. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1. iv. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.

Longer Term FIN-0022586	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.15 - No earth connection found in grey storage panel at building 03



Figure 2.16 - No equipotential bonding found at power house lighting distribution board



Figure 2.17 - No equipotential bonding found at distribution boards of building 05



Figure 2.19 - Rusted earth busbar observed without safety covers at generator room

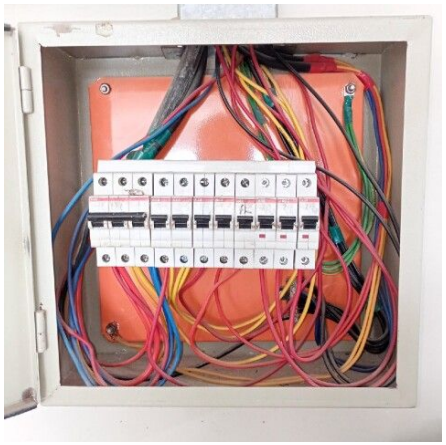


Figure 2.18 - Earth cables are inappropriately connected to the body of panel at packing area 2nd floor building 05



Figure 2.20 - Generator earth pits found having no safety covers

6.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. In a number of external areas it was observed that electrical components were not suitable for the environmental conditions in which they were installed.
ii. Several electrical panels exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements.
iii. There were visible signs of damage at electrical equipment inside LT room.

This issue was specifically observed at external wall, power house, chemical store and garment section of building 05 and facility's main gate.

The utilisation of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202505-017546

Timeline	Remediation Plan
Medium Term FIN-0023525	i. Identify and replace electrical panels, cables, and busbars that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022). ii. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials. iii. Inspect and repair or replace all damaged panels, cables, and busbars to ensure the integrity of the electrical system in accordance with Section 522 of BS7671 18th Edition + A2 (2022)
Longer Term FIN-0023526	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.21 - Damaged CT observed at main LT panel busbars in power house

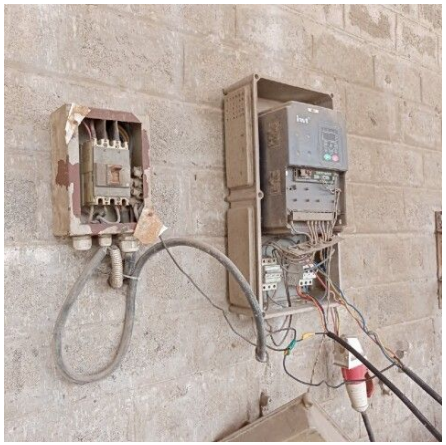


Figure 2.22 - Rusted and unprotected VFD panels installed at external wall of building 05



Figure 2.23 - Rusted and non-IP rated panel observed at chemical store building 05

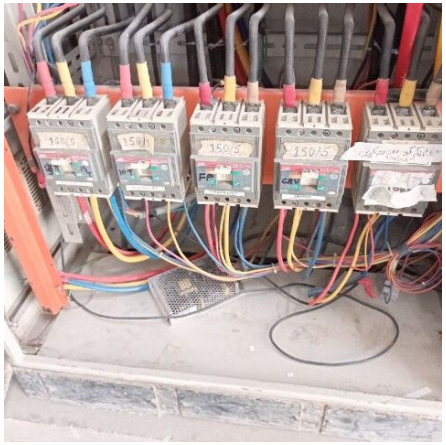


Figure 2.24 - Inappropriate placement of power supply inside DBP 01 building 05



Figure 2.25 - Inappropriate panel internal wiring observed at garment first floor building 05



Figure 2.26 - Non - IP rated panel installed at open environment at main gate

6.5 Loop Connection

This relates to the entire facility.

During the inspection loop connections were identified within the electrical panels.

In these loop connections, one circuit breaker is energised by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system. This issue was specifically observed at the first-floor entrance and at the Stenter Machine 05 in Building 05.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202505-017554

Timeline	Remediation Plan
Medium Term FIN-0022721	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilised to provide each circuit with the required live, neutral, and earth connections.



Figure 2.27 - Loop connection found at Stentier machine 05 building 05



Figure 2.28 - Loop connection found at 1st floor entrance at building 05

6.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker. This issue was specifically observed at the outdoor area of Building 05, stitching 4 office panel, DBP 03, DBP 01, and the DB at the chemical store in Building 05.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202505-017552

Timeline	Remediation Plan
Medium Term FIN-0022702	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022)



Figure 2.29 - Multiple connection observed at DB chemical store building 05



Figure 2.31 - Multiple connection observed at DBP 03 building 05



Figure 2.30 - Multiple connection observed at DBP 01 building 05

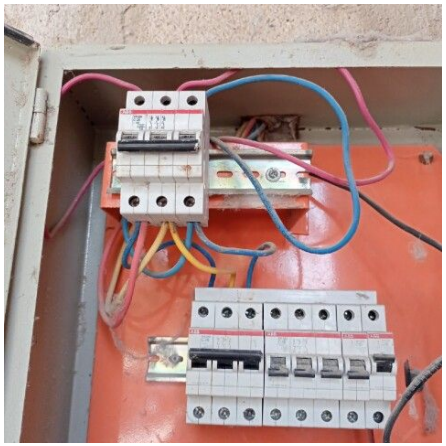


Figure 2.32 - Multiple connection observed at outdoor building 05

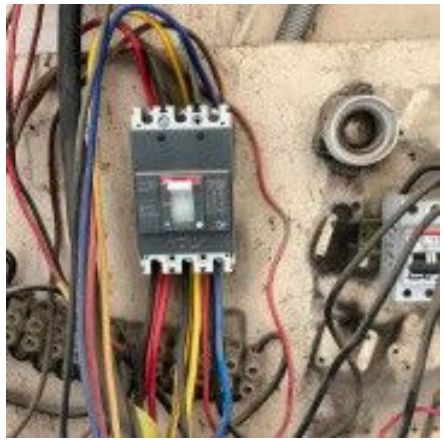


Figure 2.33 - Multiple connection observed at stitching 4 office panel

6.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings were discovered in the panels and distribution boards at processing area building 05.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202505-017553

Timeline	Remediation Plan
Medium Term FIN-0022705	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term FIN-0022706	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.34 - Opening observed at DBP 05 processing area building 05

6.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At numerous locations in power house, it was found that the electrical panels do not possess the required space for inspection or easy access to the panels and their components.

In several instances, the placement of electrical panels adjacent to machines that continuously reject heat has caused the surrounding temperature to exceed ambient levels by several degrees. This temperature rise further increases the temperature of the panels and boards nearby.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202505-017551

Timeline	Remediation Plan
Medium Term FIN-0022688	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.35 - Inadequate accessibility to control panel at power house first floor

6.9 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels and trunking systems.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

This issue was specifically observed at Stitching Unit 03, the Garment Washing Shed in Building 05, Stenter 01 in Building 05, the Grey Building 03, the Motor Control Panel in Building 05, and the Garment Section on the first floor of Building 05.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202505-017556

Timeline	Remediation Plan
Medium Term FIN-0022734	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022)
Longer Term FIN-0022735	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.36 - Lint & dust observed at grey building 03



Figure 2.37 - Lint & dust observed at Stenter 01 building 05



Figure 2.38 - Lint & dust observed at motor control panel at ground floor building 05

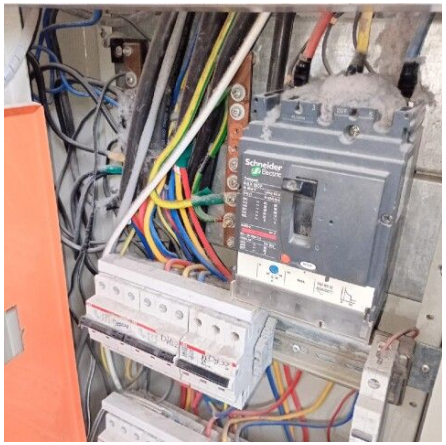


Figure 2.39 - Lint & dust observed at garment washing shed building 05



Figure 2.40 - Lint & dust observed at garment section first floor building 05



Figure 2.41 - Lint & dust observed at stitching unit 03

6.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

No lightning protection system was installed, and there was no risk assessment provided to confirm whether it was not required.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202505-017550

Timeline	Remediation Plan
Medium Term FIN-0022678	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term FIN-0022679	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning).

7. Fire Alarm & Detection System (FADS)

7.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, it was observed that there were three conventional Fire Alarm and Detection System (FADS) panels installed in the facility, covering all six buildings. However, the system was found non-listed, non-addressable, and used non-fire-rated cabling. Additionally, detection coverage was inadequate in critical areas, and several components—including sounders, MCPs, and detectors—were found faulty. These deficiencies pose significant risks to occupant safety and indicate major non-compliance with standard fire safety requirements.

The following observations were identified with the FADS:

• Detectors were not installed at the highest points in certain areas, reducing their effectiveness during fire incidents.

• Inadequate detector coverage was observed, particularly in areas such as Stitching 03 where beam detectors provided insufficient coverage.

• Detectors were missing in several critical areas, including above cable and mercerizing machines, the transformer room, the 2nd and 3rd floors of Building 05, and above the Stenter machine.

• A faulty sounder/flasher unit was observed at the power house building, which compromises audible and visual alerts during an emergency.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm.F-202505-017563

Timeline	Remediation Plan
Immediate FIN-0022770	Ensure the functionality and performance of the existing alarm and detection system within the building.

Medium Term FIN-0022771	(1) Submit design documentation for new replacement FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term FIN-0022772	(2) Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier. (3) Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation. (4) Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review. (5) Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation. (6) Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.42 - Conventional fire alarm panel installed covering all buildings



Figure 2.43 - No detectors found installed at transformer room



Figure 2.44 - No detector found installed above cable and mercerizing machine building 05



Figure 2.45 - No detector found installed above Stenter machine building 05



Figure 2.46 - No detector found installed at 2nd floor building 05



Figure 2.48 - Detector are not placed at highest point in online stores room building 05



Figure 2.47 - No detector found installed at 3rd floor building 05



Figure 2.49 - Less coverage in stitching 03 area by means of beam detectors



Figure 2.50 - Faulty sounder flasher observed at Power house building

8. Emergency Lighting

8.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency lighting was provided but not functional in a number of locations. This issue was specifically observed at the 3rd Floor Garments Section in Building 05 and the Ground Floor of Building 03.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202505-017558

Timeline	Remediation Plan
Medium Term FIN-0022750	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term FIN-0022751	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.51 - Non functional emergency light at 3rd floor garments section building 05



Figure 2.52 - Non functional emergency light at ground floor building 03

8.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

Several areas lack sufficient emergency lighting coverage. This issue was specifically observed at the main entrances of Stitching 2 and 3, between the doors of Stitching 03, the first floor of Building 05, and the carton storage area.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202505-017560

Timeline	Remediation Plan
Medium Term FIN-0022757	Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Install additional emergency luminaires in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation. Submit emergency light design layout to Pakistan Accord for approval.
Longer Term FIN-0022758	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.53 - No emergency lights installed at first floor building 05



Figure 2.54 - Insufficient emergency light at stitching 2 and 3 main entrance



Figure 2.55 - No emergency light installed at carton storage area



Figure 2.56 - No emergency light installed between doors of stitching 03

8.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages.Further the emergency lights and exit signs be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202505-017561

Timeline	Remediation Plan
Medium Term FIN-0022759	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately when general lighting power fails, as required by the Building Code of Pakistan Fire Safety Provisions 2016.
Longer Term FIN-0022760	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.57 - Emergency lighting was found not on separate circuit

9. Generator, Transformer & Sub-Station Room

9.1 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

Several battery terminals were exposed and not properly insulated in generator room, increasing the risk of electrical shock or short circuits.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202505-017564

Timeline	Remediation Plan
Medium Term FIN-0022773	Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term FIN-0022774	Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.



Figure 2.58 - Bare and corroded battery terminals observed at generator room building 05

10. Cabling & Wiring

10.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

Proper supports were not provided for cables; they were laid directly on the ground and hung along the walls. Additionally, several locations had open and exposed electrical connections. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure. This issue was specifically observed at the pump area near the main gate, stitching 4 office, batch rotation area near the boiler, and the grey department in Building 03.F-202505-017541

Timeline	Remediation Plan
Medium Term FIN-0022485	Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations)
Longer Term FIN-0022486	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.59 - Unsupported cable observed at stitching 4 office



Figure 2.60 - Unsupported cable observed at grey department building 03



Figure 2.61 - Unsupported cable observed at batch rotation area near boiler



Figure 2.62 - Unsupported cable observed at pump area main gate

10.2 Cable trench and channel cleanliness

This related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of fluff and dust, which poses a serious fire risk. This issue was specifically observed at building 05 and stitching 02.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimise fire risks associated with accumulated debris.F-202505-017544

Timeline	Remediation Plan
Medium Term FIN-0022564	Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Longer Term FIN-0022565	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.63 - Lint & dust observed at building 05 cable tray



Figure 2.64 - Lint & dust observed at stitching 2 cable trays

10.3 Splicing of cables

This relates to the whole facility.

During the inspection the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was specifically observed at open wiring on the first floor of Building 05, Strainer 01 in Building 05, the cutting machine in Stitching 02, the power house external wall, and the cutting area in Stitching 02.
ii. Machine power wires were inserted directly into power receptacles without a power plug. This practice is highly dangerous as it can lead to loose connections, electrical shocks, and short circuits. Direct insertion bypasses proper safety mechanisms, significantly increasing the risk of fire and electrical hazards, thereby endangering life safety. This issue was specifically observed at the mercerize section in Building 05, Arioli Machine 2 in Building 05, and the engraving area on the first floor.
iii. Inappropriate and exposed cable connectors were observed, posing potential electrical safety hazards and indicating non-compliance with standard wiring practices. This issue was specifically observed at DBP 08 and DBP-2 in Building 05, and at the engraving area on the first floor of Building 05, where exposed PVC connectors were also noted.

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilised. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202505-017542

Timeline	Remediation Plan
Medium Term FIN-0022506	i. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring. ii. Replace the directly inserted power wires with proper power plugs to ensure safe and secure connections. iii. Replace all inappropriate and exposed cable connectors with properly insulated and enclosed connectors in accordance with standard electrical safety codes to ensure safe and compliant terminations.

Longer Term FIN-0022507	i. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques. ii. Develop and enforce a facility-wide electrical safety protocol that mandates the use of power plugs for all equipment. Conduct regular safety audits and provide training sessions to staff to reinforce the importance of using proper power plugs and adherence to electrical safety standards. iii. Ensure all cable connectors are properly insulated and enclosed to meet electrical safety standards, replacing any exposed or inappropriate connections for secure and compliant installations.



Figure 2.65 - Spliced cable with PVC tape joint at power house external wall



Figure 2.66 - Directly inserted wires observed at mercerize at building 05



Figure 2.67 - Spliced cable observed at open wiring 1st floor building 05

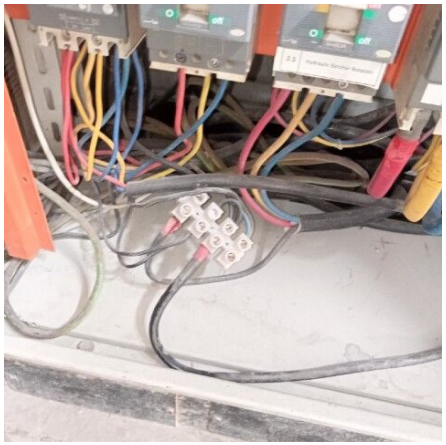


Figure 2.68 - Inappropriate cable connector observed at DBP-2 building 05



Figure 2.69 - Spliced cable with PVC tape joint at Stenter 01 building 05



Figure 2.70 - Directly inserted wires observed at Arioli machine 2 building 05



Figure 2.71 - Inappropriate cable connector observed at DBP 08 building 05



Figure 2.72 - Directly inserted wires observed at engraving area 1st floor



Figure 2.73 - Inappropriate and exposed PVC connector observed at engraving 1st floor building 05



Figure 2.74 - Spliced cable with PVC tape joint at cutting area at stitching 02



Figure 2.75 - Spliced cable and damaged insulation observed at cutting machine stitching 02

10.4 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

This issue was observed at grey storage building 03.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202505-017543

Timeline	Remediation Plan
Medium Term FIN-0022554	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term FIN-0022555	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.

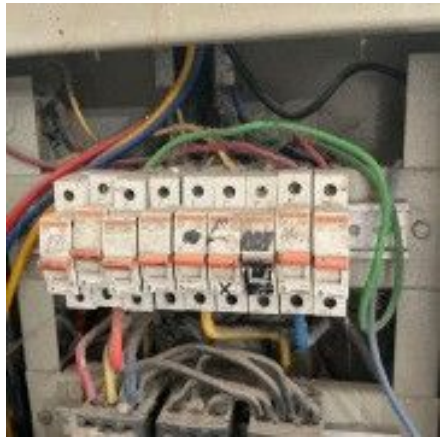


Figure 2.76 - Green cable used in phase at grey storage building 03

10.5 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles mounted on combustible wooden ply: Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety. This issue was specifically observed at the carpenter room Stitching 04 and the processing lab Building 05.

ii. Broken and exposed power receptacles: Damaged receptacles with exposed parts can lead to loose connections and increased electrical resistance. This condition can cause overheating, potentially igniting surrounding materials and increasing the risk of electrical fires, posing a serious life safety threat. This issue was specifically observed at the power house control room, processing lab Building 05, and Regani Machine 01 Building 05.F-202505-017548

Timeline	Remediation Plan
Medium Term FIN-0022613	i. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk. ii. Replace or repair broken and exposed power receptacles immediately to restore safe and secure connections.
Longer Term FIN-0022614	i. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance. ii. Implement a preventive maintenance program that includes periodic checks and timely replacement of damaged electrical components to maintain safety standards.



Figure 2.77 - Wooden backing circuit breaker observed at processing lab building 05



Figure 2.79 - Damaged fixture observed at power house control room



Figure 2.78 - Wooden fixture observed at carpenter room stitching 04

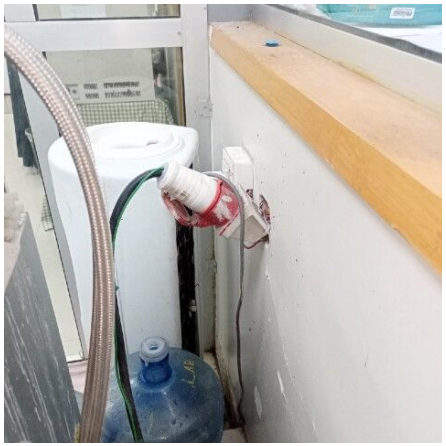


Figure 2.80 - Damaged power socket at processing lab building 05



Figure 2.81 - Wooden backing on fixtures observed at processing lab building 05



Figure 2.82 - Damaged fixture observed at Regani Machine 01 building 05

11. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.